

Bubble Sort

Definition

Bubble Sort is a sorting algorithm that repeatedly compares **two adjacent elements** and swaps them if they are in the wrong order. After each pass, the largest unsorted element moves to the end of the array.

Algorithm

1. Start from the first element.
2. Compare the first and second elements.
3. If the first element is greater than the second, swap them.
4. Move to the next pair of adjacent elements.
5. Continue until the end of the array.
6. Repeat the process for the remaining unsorted elements.
7. Stop when all elements are sorted.

Output:

```
sorted array: [11, 12, 22, 25, 34, 64, 90]  
...Program finished with exit code 0  
Press ENTER to exit console.█
```

Time Complexity

- **Best Case:** $O(n)$ — when the array is already sorted (optimized Bubble Sort).
- **Average Case:** $O(n^2)$
- **Worst Case:** $O(n^2)$

Space Complexity

- **$O(1)$** — Bubble Sort sorts the array in-place and uses no significant extra memory.

